electrons can never produce enough degeneracy pressure to counterbalance the star's gravity[4]; the star must, in Arthur Eddington's words, "go on radiating and radiating and contracting and contracting, until, I suppose, it gets down to a few kilometers radius, when gravity becomes strong enough to hold in the radiation, and the star can at last find peace."

Or that would be its fate, if not for neutron stars. If Zwicky was right that neutron stars can exist, then they must be rather analogous to white-dwarf stars, but with their internal pressure produced by neutrons instead of electrons. This means that there must be a neutron-star curve in Figure 5.3, analogous to the white-dwarf curve, but at circumferences (marked on the horizontal axis) of roughly a hundred kilometers, instead of tens of thousands of kilometers. On this neutron-star curve neutron pressure would balance gravity perfectly, so neutron stars could reside there forever.

Suppose that the neutron-star curve extends upward in Figure 5.3 to large masses; that is, suppose it has the shape labeled *B* in the figure. Then Sirius, when it dies, *cannot* create a black hole. Rather, Sirius will shrink until it hits the neutron-star curve, and then it can shrink no more. If it tries to shrink farther (that is, move to the left of the neutron-star curve into the shaded region), the neutrons inside it will protest against being squeezed; they will produce a large pressure (partly due to degeneracy, that is, "claustrophobia," and partly due to the nuclear force); and the pressure will be large enough to overwhelm gravity and drive the star back outward. If the star tries to reexpand into the white region, the neutrons' pressure will decline enough for gravity to take over and squeeze it back inward. Thus, Sirius will have no choice but to settle down onto the neutron-star curve and remain there forever, gradually cooling and becoming a solid, cold, black neutron star.